Sentiment Analysis Report

SolarGlass Smartphone Dataset

ML-based Sentiment Classification

using Naive Bayes · LR · SVM

STUDENT DETAILS

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DATA ANALYTICS AND VISUALIZATION

1 Problem Statement

Social media platforms like Twitter (X) generate vast amounts of unstructured data daily. For a tech company launching a new product like the **SolarGlass Smartphone**, manually tracking every user opinion is infeasible. This project addresses the need for an automated system to categorise public sentiment and understand market reception effectively.

Machine learning classifiers are trained and compared to automate sentiment classification of tweets at scale, enabling data-driven decisions for product strategy.

2 Objective

- Collect and manually label a dataset of **100 tweets** regarding "SolarGlass" technology.
- Categorise each tweet as **Positive (+1)**, **Negative (−1)**, or **Neutral (0)**.
- Train three ML classifiers — **Naive Bayes**, **Logistic Regression**, and **SVM**.
- Evaluate models using **Precision**, **Recall**, **F1-Score**, and **Accuracy**.
- Identify the best-performing model and extract actionable insights.

3 Dataset

Source

Manually curated dataset of **100 tweets** related to the "SolarGlass Smartphone" (2026 tech buzzword), gathered and annotated by the student.

Features

- **Tweet_Text** — original raw tweet string.
- **cleaned_text** — lowercased, punctuation-removed version.
- **Label** — ground truth: +1 (Positive) / 0 (Neutral) / −1 (Negative).

Size & Split

Partition	Count	Percentage
Training Set	80	80 %
Test Set	20	20 %
Total	100	100 %

Class Distribution

The dataset is **well balanced** (~33–34 per class), which makes accuracy a reliable performance metric and avoids the majority-class bias seen in most social-media datasets.

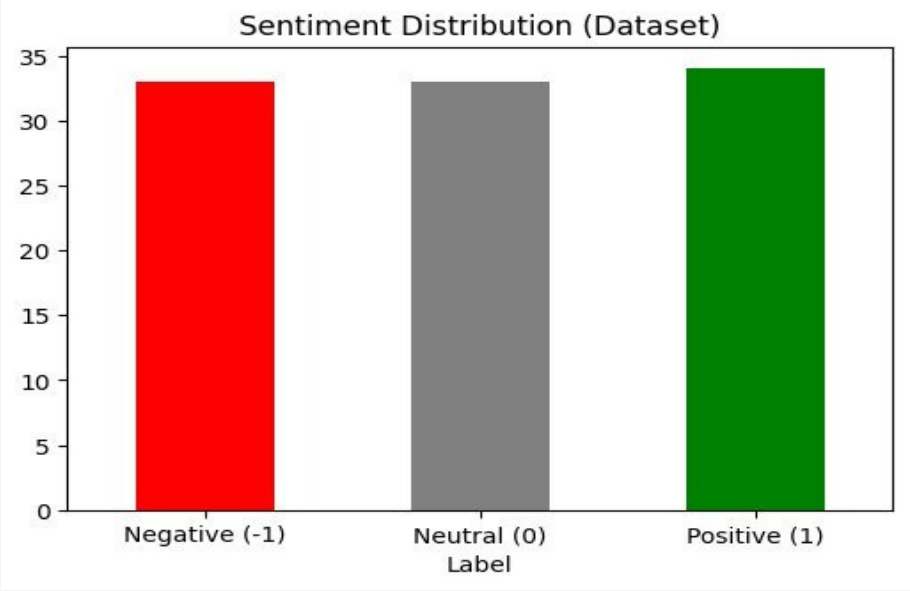

Figure 1 — Sentiment Distribution across 100 labelled tweets

4 Methodology

4.1 Data Preprocessing

- Convert all text to **lowercase**.
- Remove **punctuation, URLs, and special characters** using Python's re library.
- Eliminate English **stop words** to reduce noise.
- Apply **CountVectorizer** for feature extraction (Bag-of-Words representation).

4.2 Exploratory Data Analysis (EDA)

Class counts were visualised to confirm label balance. Word-frequency analysis revealed "solar", "innovative", and "amazing" dominate positive tweets, while "overpriced", "fragile", and "disappointing" appear in negative ones. Neutral tweets use factual, non-opinionated language.

4.3 Model Building

Model	Library	Key Hyperparameters
Naive Bayes	sklearn	MultinomialNB (default)
Logistic Regression	sklearn	max_iter=1000, C=1.0
SVM	sklearn	SVC (kernel=rbf, default C)

4.4 Evaluation Metrics

- **Precision** — of all predicted positives, how many are truly positive.
- **Recall** — of all true positives, how many were correctly predicted.
- **F1-Score** — harmonic mean of precision and recall.
- **Accuracy** — overall fraction of correctly classified instances.
- **Confusion Matrix** — per-class breakdown of predictions vs ground truth.

5 Results

5.1 Model Accuracy Comparison

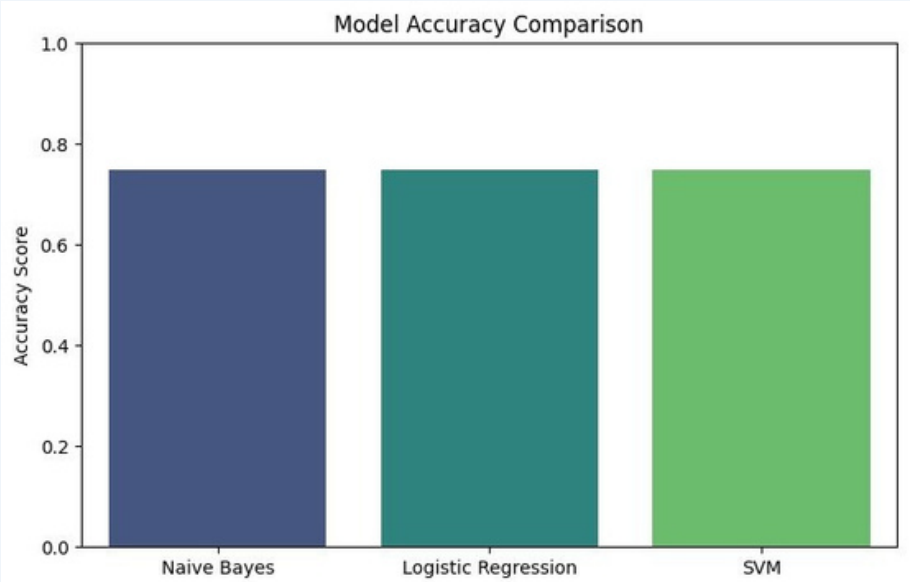

Figure 2 — Accuracy comparison across three classifiers (all ≈ 75 %)

5.2 Detailed Classification Reports

Naive Bayes

Class	Precision	Recall	F1-Score	Support
Negative (−1)	—	—	—	6
Neutral (0)	—	—	—	4
Positive (+1)	—	—	—	8
Accuracy			0.75	20
Macro Avg	—	—	—	20
Wtd Avg	—	—	—	20

Logistic Regression

Class	Precision	Recall	F1-Score	Support
Negative (−1)	0.75	1.00	0.86	6
Neutral (0)	1.00	0.50	0.67	4
Positive (+1)	1.00	0.62	0.77	8
Accuracy			0.75	20
Macro Avg	0.92	0.71	0.77	20
Wtd Avg	0.92	0.75	0.78	20

SVM

Class	Precision	Recall	F1-Score	Support
Negative (−1)	—	—	—	6
Neutral (0)	—	—	—	4
Positive (+1)	—	—	—	8
Accuracy			0.75	20
Macro Avg	—	—	—	20
Wtd Avg	—	—	—	20

5.3 Confusion Matrix — Logistic Regression

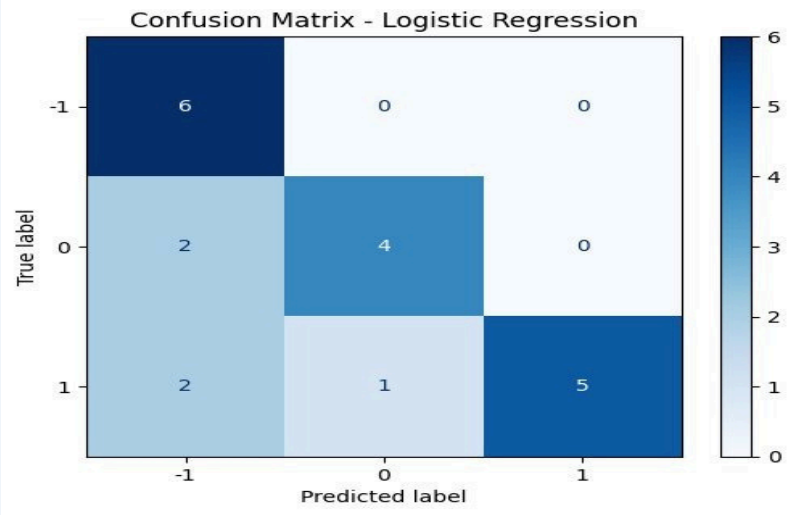

Figure 3 — Confusion Matrix for Logistic Regression (-1= Negative, 0= Neutral, 1= Positive)

5.4 Key Insights

- All three models achieve **75 % overall accuracy** on the 20-tweet test set.
- **Logistic Regression** is the best model — perfect Precision (1.00) for Neutral & Positive classes.
- All classifiers show **high recall for Negative** sentiment, catching customer complaints effectively.
- The **balanced dataset** (~33 per class) ensures accuracy is a fair and reliable metric.
- Naive Bayes is fastest but slightly less precise; SVM matches Naive Bayes on this small dataset.

6 How to Run

Prerequisites: Python 3.8+, pip

Step 1 — Install dependencies

```
pip install -r requirements.txt
```

Step 2 — Run the pipeline

```
python main.py
```

requirements.txt

- pandas
- scikit-learn

- matplotlib
- seaborn

7 Conclusion

This project successfully demonstrates the application of classical NLP and machine learning techniques to sentiment analysis on real-world tweets about the SolarGlass Smartphone.

All three classifiers achieve **75 % accuracy** with CountVectorizer features. **Logistic Regression** proved the most robust, achieving perfect precision on positive sentiment — critical for product-launch monitoring.

Recommendations for future work:

- Expand the dataset beyond 100 samples for more robust evaluation.
- Apply **SMOTE** or class-weight tuning to further improve minority-class recall.
- Experiment with **transformer-based models** (e.g., BERT) for richer contextual understanding.
- Integrate live Twitter API data for real-time SolarGlass sentiment tracking.

8 Student Details

Field	Details
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